System-level scheduling of rotating-chamber semiconductor cluster tools: A source-configurable two-stage MILP and structure-aware exact search

Lizhen Shao[a,*], Lincong Wei[a]

[a]*Department of Control Science and Engineering, University of Science and Technology Beijing, No. 30 Xueyuan Road, Haidian District, Beijing 100083, China*

Abstract

Rotating-chamber cluster tools are equipment-level manufacturing systems in which wafer flow, chamber rotation, cleaning, reusable dummy wafers, and shared robots must be coordinated without intermediate buffering. This study develops a source-configurable scheduling framework that separates external material interfaces from a common internal configuration–resource–timing model. A single-source tool is the base specialization; a dual-source mixed-flow $4\times1/2\times2$ configuration is used as the strongly coupled benchmark. The finite-horizon two-stage mixed-integer linear program first minimizes end-to-end makespan and then preserves the selected discrete schedule and makespan within numerical tolerance while refining physical route waits and chamber cadence. Exact solution is enhanced by a solver-checked structure-preserving binary-skeleton (SPBS) warm start and RDMCT-HBS, which combines variable-role cut features, hierarchical learning, bounded beam decoding, and structural cut completion inside SCIP. Four exactly solved cases check internal model consistency. Across 1500 equal-600-s workflow runs, the two-stage workflow reduces total route wait, maximum route wait, cadence variation, and cadence deviation by 51.91%, 68.89%, 39.70%, and 94.05%, respectively. In 2700 nominal runs sharing the same SPBS incumbent, RDMCT-HBS records the lowest mean primal–dual integral and solution time among seven solver configurations. A separate 960-run paired com-

*Corresponding author.

Email addresses: lshao@ustb.edu.cn (Lizhen Shao), m202420836@xs.ustb.edu.cn (Lincong Wei)

parison associates SPBS with 26.19% lower mean PDI and increases incumbent availability from 51.67% to 100%. All 729 larger shifted-instance runs return solver-checked feasible schedules. The results demonstrate system-level operational gains while retaining exact-solver feasibility and bound semantics.

Keywords: semiconductor manufacturing, cluster tool, scheduling, mixed-integer linear programming, branch-and-cut, warm start

1. Introduction

Semiconductor equipment is capital intensive, and its effective capacity is often determined by scheduling rather than nominal processing speed. Cluster tools illustrate this dependence particularly clearly: chambers, load locks, and robotic handlers form a tightly coupled system in which a locally sensible decision can delay several downstream operations. Better scheduling therefore offers a direct route to higher asset utilization without additional hardware (Chien and Lan, 2021; Hong et al., 2026).

This paper studies a class of rotating-chamber cluster tools used in a plasma-enhanced chemical vapor deposition setting. A configurable source interface connects external release and return domains to a common internal network. With one interface, the model represents a single-source tool. The evaluated dual-source extension sends product wafers from atmospheric load ports and reusable dummy wafers from vacuum-zone storage; both streams merge inside rotary chambers and compete for vacuum transport. Two recipe families coexist: $4{\times}1$ operation processes two chamber sides as a rotary batch, whereas 2×2 operation links consecutive product pairs through overlapping exposures. Cleaning creates dummy-only operations and splits processing into epochs. Scheduling must therefore determine active cycles, cleaning boundaries, physical dummy identities, every conflicting robot and module order, and continuous event times. In the evaluated instances, same-mode wafers are paired by identifier and mapped to fixed canonical chamber–side slots before optimization. Pairing, chamber, and side are fixed instance conventions rather than optimized decisions.

Existing cluster-tool research provides important models for cyclic operation, multiple wafer types, residency, cleaning, and multi-space modules (Qiao et al., 2022; Li et al., 2022; Wang et al., 2023a; Li and Yang, 2025; Yang et al., 2025; Gu et al., 2024). Those models do not jointly represent the

finite-horizon dual-source flow, mixed rotary recipes, reusable dummy inventory, and shared transport considered here. The difference is structural. A prescribed robot sequence can be evaluated by an LP, but the present equipment still requires activation, reusable-dummy assignment, cleaning, resource ordering, and event times to be determined together for the fixed canonical recipe layout.

Learning-based scheduling offers powerful search guidance, but most studies act directly on dispatching or process control (Liu et al., 2022; Zhang et al., 2024; Kim and Lee, 2025; Kim and Kim, 2025). The closely related study of Hong et al. (2026) uses DDPG and discrete-event simulation to adjust continuous processing times in a PVD cluster tool. Our problem is different: physical processing times are given, while the method must construct a feasible discrete schedule and retain valid lower bounds and optimality proofs. This motivates learning inside an exact solver rather than replacing the solver.

The paper resolves two linked gaps. The modeling gap is addressed by a two-stage MILP that separates throughput from schedule quality without separating the physical system. The computational gap is addressed by embedding equipment structure in both primal initialization and cut selection. The resulting research logic is direct: rotating-chamber manufacturing physics defines the MILP; the MILP exposes its search bottlenecks; and those bottlenecks define the information used by the learning-enhanced branch-and-cut method.

The contributions are threefold.

1. We formulate a source-configurable finite-horizon MILP for rotating-chamber scheduling under canonical same-mode pairing and slot mapping. It separates external material interfaces from common internal decisions and jointly determines cycle and operation activation, rotary events, robot transfers, load-lock states, cleaning epochs, residency, reusable dummy-wafer allocation, resource order, and event timing. The dual-source mixed-flow case serves as the strongly coupled computational benchmark.

2. We introduce a makespan-preserving second stage. Unlike makespan-only scheduling, it explicitly reshapes physical product waits and chamber cadence within the discrete schedule retained from stage one.

3. We develop RDMCT-HBS, a structure-aware enhancement of HEM that

couples a feasible binary-skeleton warm start, variable-role cut features, A3C-inspired parallel learning, bounded beam decoding, and structural set completion while leaving feasibility and proof to SCIP.

The evidence design links the contributions to complementary tests. Exact solves and post-solve checks assess internal formulation consistency; the nominal comparison measures algorithmic performance; matched contrasts compare warm-start and refinement workflows; and factorial and shifted-instance studies characterize computational difficulty and feasible-schedule delivery.

2. Research background and positioning

2.1. Cluster-tool scheduling models

Cluster-tool models have progressed from steady-state robot cycles to richer representations of product heterogeneity and auxiliary constraints. Integrated patterns and workload balance have been investigated for dual-arm tools and multiple wafer types (Kim et al., 2021; Ko et al., 2021; Zhu et al., 2022). Recent MIP and analytical studies coordinate release order, robot tasks, chamber assignment, and residency for concurrent or multi-type processing (Li and Yang, 2025; Yang et al., 2025). Cleaning, purge, and post-processing constraints further demonstrate that temporal feasibility cannot be inferred from chamber capacity alone (Qiao et al., 2022; Lee et al., 2023; Yuan et al., 2023; Zhu et al., 2023; Pan et al., 2024).

Rotating and multi-space modules introduce side-dependent loading and shared chamber motion. Gu et al. (2024) analyse multi-finger robotic tools with multi-space process modules, providing the closest physical foundation for the chamber mechanism studied here. Their predetermined cyclic sequences do not cover finite mixed batches or a second reusable material stream. The present problem extends the decision boundary beyond timing a prescribed robot sequence to jointly selecting cycle activation, dummy identities, cleaning epochs, resource orders, and event times. Same-mode product pairs and chamber–side slots remain fixed by the canonical instance construction.

2.2. Learning-enhanced scheduling and exact optimization

Deep reinforcement learning has been combined with genetic, adaptive, and problem-specific search in semiconductor scheduling (Chien and Lan,

2021; Liu et al., 2022; Zhang et al., 2024; Li and Chang, 2022). These methods show that learned policies are most useful when coupled with domain structure. Exact optimization adds a complementary advantage: explicit constraints, primal and dual bounds, and verifiable feasibility. Recent learning-enhanced MIP research guides branching, primal search, and cut selection without surrendering those properties (Bengio et al., 2021; Khalil et al., 2022; Scavuzzo et al., 2024).

Cut selection is especially relevant because valid inequalities differ in both immediate efficacy and the decisions they constrain. Reinforcement learning, supervised ranking, look-ahead imitation, and adaptive scoring have all improved cut management (Tang et al., 2020; Huang et al., 2022; Paulus et al., 2022; Turner et al., 2023). HEM represents selection as a hierarchical count-and-sequence decision (Wang et al., 2023b, 2024). However, generic cut features cannot distinguish an inequality acting mainly on discrete activation from one acting on continuous event timing; greedy decoding also discards alternative partial sequences. These limitations are particularly consequential in the proposed MILP because discrete equipment choices and continuous timing are strongly coupled.

Table 1 summarizes the research position. The novelty is not the isolated use of MILP, reinforcement learning, or beam search. It is the integration of a source-configurable rotary scheduling model with a solver method whose primal and dual guidance are both derived from that model's structure.

3. Industrial problem and two-stage MILP

3.1. System logic and material interaction

Let $\mathcal{S}$ be the set of configured material-source interfaces and let $\chi_{is} \in \{0,1\}$ be fixed incidence indicating whether material unit i is associated with source s, with $\sum_{s\in\mathcal{S}} \chi_{is} = 1$. If $\mathcal{A}^{\text{int}}$ denotes the internal process and transfer arcs and $\mathcal{A}_s$ contains the release and return arcs of source s, then the complete route network is $\mathcal{A} = \mathcal{A}^{\text{int}} \cup (\bigcup_{s\in\mathcal{S}} \mathcal{A}_s)$. Changing from one configured source to two changes only the source incidence and interface arcs; rotary decisions, internal resource conflicts, and event timing retain the same variable and constraint families. The formal computational study uses the two-source specialization because it activates the strongest material-interface and shared-resource coupling.

Figure 1 shows the system at the level needed to understand the optimization model. Product wafers traverse the atmospheric domain, upper

Table 1: Research positioning relative to representative cluster-tool studies.

Study	Equipment setting	Decision scope	Solution approach
Qiao et al. (2022)	Cleaning-constrained tool	Residency and cleaning	Binary IP
Gu et al. (2024)	Multi-space rotary modules	Timing of preset sequences	LP
Yang et al. (2025)	Two wafer types	Release, robot tasks, residency	MIP
Kim and Lee (2025)	Concurrent cluster tool	Release and robot sequence	DRL + search
Hong et al. (2026)	PVD cluster tool	Continuous process-time control	DES + DDPG
This study	Source-configurable rotary tool; dual-source benchmark	Activation, dummy allocation, resources, cleaning, and event times under canonical slots	Two-stage MILP + learned B&C

load lock, vacuum robot, rotary chamber, lower load lock, and return path. Dummy wafers begin and end in the vacuum domain. The two streams share the vacuum transfer robot (VTR) and chambers but have different release and return requirements; the atmospheric transfer robot (ATR) handles the external product-wafer route.

The recipe determines how the flows interact. In 4×1 mode, the chamber loads a front side, rotates, loads a back side, processes both sides, and unloads them in reverse order. A product-deficient final side is completed by dummy wafers. In 2×2 mode, product pairs form a chain: adjacent exposures share an intermediate pair, while leading and trailing positions require dummy pairs. Cleaning interrupts the chain only at a legal exposure boundary and uses a dummy-only chamber cycle. A physical dummy wafer cannot be assigned to overlapping chamber occupancies.

This interaction creates three decision layers. The *configuration layer* chooses active cycles, cleaning epochs, and physical dummy wafers within the fixed canonical pairing and slot map. The *resource layer* orders all operations competing for robots, load locks, and chambers. The *timing layer* assigns event times and enforces residence limits. These layers must be optimized jointly because activation and dummy reuse change both resource conflicts

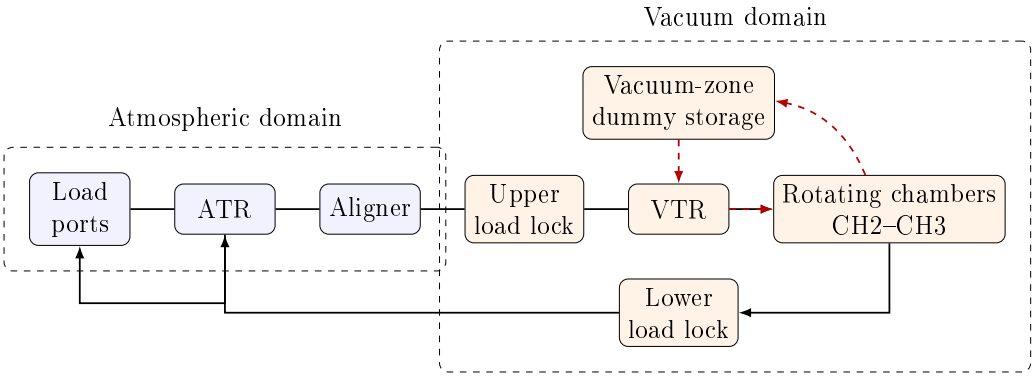


Figure 1: Dual-source material flows and shared resources. Solid arrows denote the product-wafer route; dashed arrows denote reusable vacuum-zone dummy wafers.

and the feasible timing network.

3.2. Decision structure and principal constraints

Let $\mathcal{W}$ denote product wafers, $\mathcal{P}$ the same-mode product pairs formed by the canonical preprocessing rule, $\mathcal{C}$ the rotary chambers, $\mathcal{K}$ the ordered candidate recipe slots, $\mathcal{O}$ the candidate operations, and $\mathcal{R}$ the unit-capacity resources. $\mathcal{K}(p)$ is the set of slots compatible with pair p; $\mathcal{K}^- \subseteq \mathcal{K}$ contains the nonterminal slots, and $\mathrm{succ}(k)$ is the next slot in the same ordered slot family; $\mathcal{A}$ is the technological-arc set, and $\mathcal{O}_r$ contains the operations that may use resource r. For chamber c, $\mathcal{J}_c^{\mathrm{DW}}$ is the set of dummy-wafer demands, $\mathcal{T}_c$ its physical dummy inventory, $\mathcal{V}_c$ its candidate process cycles, and $\mathcal{E}_c$ its ordered cleaning epochs. The fixed incidence $\bar{x}_{pk} \in \{0,1\}$ encodes the canonical pair-to-slot map. The binary solver variable x_{pk} is fixed to this incidence and retained for compatibility with downstream activation links.

Binary variables u_k, b_o, and b_v activate slots, operations, and candidate process cycles, respectively; a_{vce} assigns active cycle v to epoch e; z_{ce} activates an epoch; $g_{j\tau}$ assigns dummy demand j to physical dummy τ; and $y^r_{oo'}$ orders two operations sharing resource r. The activation q_j of dummy demand j is linked to its generating cycle. Each operation o has start s_o, completion e_o, active duration $d_o b_o$, and fixed duration $d_o > 0$. Constants $\delta_{oo'}$ and $\rho^r_{oo'}$ are technological and endpoint-dependent setup separations, and M is a scheduling-horizon bound. For readability, the following equations state the

principal families rather than every event-specific route and rotary equality:

$$\sum_{k\in\mathcal{K}(p)} \bar{x}_{pk} = 1 \qquad p \in \mathcal{P}, \tag{1}$$

$$x_{pk} = \bar{x}_{pk} \qquad p \in \mathcal{P},\ k \in \mathcal{K}(p), \tag{2}$$

$$x_{pk} \le u_k \qquad p \in \mathcal{P},\ k \in \mathcal{K}(p), \tag{3}$$

$$u_{\mathrm{succ}(k)} \le u_k \qquad k \in \mathcal{K}^{-}, \tag{4}$$

$$e_o = s_o + d_o b_o \qquad o \in \mathcal{O}, \tag{5}$$

$$s_{o'} \ge e_o + \delta_{oo'} - M(2 - b_o - b_{o'}) \qquad (o,o') \in \mathcal{A}, \tag{6}$$

$$s_{o'} \ge e_o + \rho^r_{oo'} - M(1 - y^r_{oo'}) - M(2 - b_o - b_{o'}), \quad r \in \mathcal{R},\ \{o,o'\} \subseteq \mathcal{O}_r, \tag{7}$$

$$s_o \ge e_{o'} + \rho^r_{o'o} - M y^r_{oo'} - M(2 - b_o - b_{o'}), \quad r \in \mathcal{R},\ \{o,o'\} \subseteq \mathcal{O}_r, \tag{8}$$

$$\sum_{\tau\in\mathcal{T}_c} g_{j\tau} = q_j \qquad c \in \mathcal{C},\ j \in \mathcal{J}_c^{\mathrm{DW}}. \tag{9}$$

Equations (1)–(2) disclose the fixed canonical slot map rather than an endogenous grouping decision. Equations (3)–(4) link that map to a contiguous active structure, while Eq. (5) makes activation consistent with event timing. Technological arcs in Eq. (6) create active product and dummy routes. The activation-qualified disjunctions (7)–(8) prevent overlap on the ATR, aligner, load-lock slots, VTR, chambers, and each reusable dummy wafer; ρ includes endpoint-dependent robot repositioning.

Table 2 connects the remaining industrial rules to their mathematical role. This mapping is important because feasibility is a property of the complete equipment trajectory, not only of chamber occupancy.

Cleaning epochs satisfy

$$\sum_{e\in\mathcal{E}_c} a_{vce} = b_v \qquad c \in \mathcal{C},\ v \in \mathcal{V}_c, \tag{10}$$

$$a_{vce} \le z_{ce}, \qquad \sum_{v\in\mathcal{V}_c} a_{vce} \le H z_{ce} \qquad c \in \mathcal{C},\ e \in \mathcal{E}_c, \tag{11}$$

$$z_{c,e+1} \le z_{ce} \qquad c \in \mathcal{C},\ e, e+1 \in \mathcal{E}_c, \tag{12}$$

where H is the maximum number of active process cycles between cleans. Linking constraints insert a dummy-only clean between consecutive active epochs and allow a 2×2 chain to split only at an exposure boundary. Residence limits take the form

$$0 \le s_{\mathrm{out}(w,r)} - e_{\mathrm{in}(w,r)} \le \overline{R}_{wr}. \tag{13}$$

Table 2: Industrial rules represented by the MILP.

Industrial rule	Mathematical mechanism	Scheduling consequence
Rotary side sequence	Event equalities and precedence arcs	Enforces load–rotate–process–unload physics
Mixed $4 \times 1/2 \times 2$ recipes	Fixed canonical map, activation, and adjacency links	Activates feasible cycles and shared exposures
Cleaning interval	Cycle-to-epoch assignment and capacity	Inserts dummy-only cleaning at legal boundaries
Reusable dummy inventory	Physical-wafer assignment plus interval disjunction	Prevents simultaneous reuse of one dummy wafer
Robot/load-lock capacity	Endpoint-dependent resource ordering	Prevents transfer conflicts and includes repositioning
Residence restrictions	Upper bounds between paired route events	Protects time-sensitive products and transfers
Finite-batch throughput	Latest return-to-load-port variable	Measures end-to-end rather than chamber-only completion

In cluster-tool control, deadlock is a reachable equipment state in which wafer–resource dependencies form a circular wait; Petri-net formulations address this controller-level reachability property explicitly (Nishi and Matsumoto, 2015). The present finite-horizon MILP makes a narrower, offline-schedule claim. Every active route has a finite source-to-return precedence chain, and every active pair sharing a unit-capacity resource receives one temporal order. Provided all physical hold intervals and transfers are represented and active separations are positive, a directed cycle in the combined route–resource graph would make the summed timing inequalities imply $0 > 0$ and would therefore be infeasible. Terminal constraints require every product and assigned dummy wafer to return. Thus a feasible solution is a complete scheduled trajectory without circular wait under the modeled conditions; this is not a controller-level deadlock-freeness guarantee under dispatch deviations, unmodeled buffers, or equipment failures.

3.3. Throughput first, schedule quality second

Stage one minimizes the return time of the last product wafer:

$$C_{\max} \geq e_w^{\text{return}} \quad (w \in \mathcal{W}), \qquad \min_{\xi \in \mathcal{F}} C_{\max}(\xi), \tag{14}$$

where ξ collects the model variables and $\mathcal{F}$ contains all activation, canonical-map, rotary, cleaning, dummy-allocation, resource, and timing constraints. A makespan objective alone is insufficient operationally. Once the last return time is fixed, many schedules may be equivalent in $C_{\max}$ but differ substantially in intermediate product waiting, extreme holds, and chamber cadence. Those differences affect exposure consistency, work-in-process residence, and recovery from small disturbances.

Let $\widehat{\xi}$ be the best stage-one schedule and $\widehat{C}$ its makespan, and let $\mathcal{D}_{\text{fix}}$ index the discrete variables retained from that incumbent. Stage two imposes

$$\xi_d = \widehat{\xi}_d \ (d \in \mathcal{D}_{\text{fix}}), \qquad C_{\max}(\xi) \leq \widehat{C} + \varepsilon_C. \tag{15}$$

It then minimizes

$$\Phi(\xi) = \lambda_1 L_{\text{sum}} + \lambda_2 L_{\max} + \lambda_3 D_{\text{cadence}} + \lambda_4 E_{\text{cap}}, \tag{16}$$

where $\omega_\alpha \geq 0$ $(\alpha \in \mathcal{Q})$ are the model-native wait and idle variables, and $h_{mce,r}$ is the rth active processing start in the retained order for recipe mode m, chamber c, and cleaning epoch e. The absolute values, maximum, and

positive parts in the operational description are implemented with the following linear epigraph relations, which are tight for the positively weighted terms at an optimum:

$$L_{\text{sum}} = \sum_{\alpha \in \mathcal{Q}} \omega_\alpha, \qquad L_{\max} \geq \omega_\alpha \qquad \alpha \in \mathcal{Q}, \tag{17}$$

$$v_{mce,r} \geq (h_{mce,r+1} - h_{mce,r}) - (h_{mce,r} - h_{mce,r-1}), \qquad (m,c,e,r) \in \mathcal{I}_h, \tag{18}$$

$$v_{mce,r} \geq -(h_{mce,r+1} - h_{mce,r}) + (h_{mce,r} - h_{mce,r-1}), \quad (m,c,e,r) \in \mathcal{I}_h, \tag{19}$$

$$D_{\text{cadence}} = \sum_{(m,c,e,r) \in \mathcal{I}_h} v_{mce,r}, \qquad v_{mce,r} \geq 0, \tag{20}$$

$$\eta_\alpha \geq \omega_\alpha - \bar{\omega}_\alpha, \quad \eta_\alpha \geq 0, \qquad E_{\text{cap}} = \sum_{\alpha \in \mathcal{Q}} \eta_\alpha. \tag{21}$$

Here $\mathcal{I}_h$ contains the interior indices $r = 2, \ldots, n_{mce} - 1$ of retained chamber-start sequences with $n_{mce} \geq 3$, and $\bar{\omega}_\alpha$ is the configured soft wait cap. The nonnegative weights $\lambda_1, \ldots, \lambda_4$ set the trade-off among aggregate waiting, extreme waiting, cadence irregularity, and soft wait-cap excess. The formal linear profile uses weights $(1, 10, 25, 0)$; the last term is disabled and not interpreted when its weight is zero; it is tight when a positive soft-cap weight is configured. Physical product-route waits reported as operational outcomes are reconstructed uniformly from event times, rather than read from these objective auxiliaries; their six-segment definition is given in the supplement.

Here ε_C is only the numerical equality tolerance. Equation (15) gives the two-stage model a clear guarantee. The second stage cannot worsen the retained makespan beyond this tolerance and cannot change the chosen activation, cleaning, dummy-allocation, or resource-order structure. If stage one proves optimality, the refined schedule remains globally makespan-optimal within the same tolerance. If stage two also proves optimality, its timing is optimal for Eq. (16) within the retained discrete structure; otherwise the returned refinement is feasible under the makespan cap but is not claimed to be stage-two optimal. If stage one ends with an incumbent, makespan retention is relative to that incumbent. The model therefore separates objectives without disconnecting them from the same physical schedule.

3.4. Modeling advance

The formulation extends existing rotating or multi-space models in four directions simultaneously: finite rather than steady-state scheduling; a config-

urable source interface whose single-source base admits a dual-source mixed-flow specialization; joint activation, cleaning, dummy-allocation, resource-order, and timing decisions on a canonical mixed-recipe layout; and conditional schedule-quality refinement rather than makespan alone. This combination matters because source topology, dummy availability, cleaning, and active rotary cycles alter the robot conflict graph itself. They cannot be appended as independent capacity constraints after a complete product trajectory has been timed.

The same structure explains computational difficulty. Activation, epoch, dummy identity, and pairwise order binaries interact with a dense event time network. The resulting cuts may affect different algebraic roles even when their generic efficacy scores are similar. This observation motivates the solver design in the next section.

4. Structure-aware learning-enhanced exact solution

4.1. From model structure to solver guidance

The MILP exhibits two practical symptoms: finding the first competitive incumbent can be slow, and generic cut scores do not reveal whether a cut tightens discrete equipment choices, continuous timing, or their coupling. RDMCT-HBS addresses the symptoms at complementary points. Structure-preserving binary-skeleton (SPBS) initialization supplies primal guidance before branch-and-cut, while the learned selector supplies dual guidance during separation. Both components remain advisory: SCIP keeps the original feasible region, validates incumbents, generates valid cuts, and retains proof responsibility.

4.2. Structure-preserving binary-skeleton warm start

SPBS retains the canonical pair-to-slot equalities and fixes candidate activation, cleaning-epoch, load-lock, and selected resource-order roles in a temporary feasibility MIP. If a profile finds no incumbent, VTR, load-lock, and other order fixings are progressively released; continuous event times always remain free. A start is accepted only after reconstruction and complete feasibility checking in the full model. Only that feasible schedule—not the temporary fixing constraints—is injected into the unrestricted production model. Formal evaluations use this incumbent after it has been precomputed.

This design is stronger than an arbitrary primal hint. The retained decisions encode the combinatorial architecture that the detailed timing model

must otherwise discover from scratch, while the completion solve repairs local timing interactions. Because the original model receives only a feasible incumbent, SPBS cannot remove a better solution or weaken an optimality proof.

4.3. Variable-role reconstruction and hierarchical decoding

HEM describes candidate cuts with 13 generic statistics. We add ten structural features derived from solver metadata. Variables are partitioned into binary, general integer, objective-active continuous, auxiliary continuous, and other roles. For cut i, the coefficient mass on role g is

$$p_{ig} = \frac{\sum_{j:x_j \in g} |a_{ij}|}{\sum_j |a_{ij}| + \varepsilon}. \tag{22}$$

Role masses, bounded-variable mass, coefficient sign balance, discrete–continuous coupling, dominant-role mass, and role entropy extend the representation to 23 dimensions. The features are invariant to variable names and distinguish broad algebraic variable domains. They do not, by themselves, identify equipment semantics among variables in the same domain. Candidates are screened deterministically by efficacy to at most 128 rows, and the selector returns at most 16 cuts.

The 13 generic and 10 role features are standardized with the training statistics stored with each trained parameter set. Layer normalization, separate nonlinear projections of the two blocks, sigmoid-gated fusion, a one-layer LSTM, and one pointer-attention glimpse produce the policy state. The LSTM hidden size is 128 and the tanh exploration constant is 5. The high-level output is a continuous selection fraction capped at 0.2; its discrete cardinality is obtained by flooring the candidate count times that fraction, applying a minimum-two safeguard when feasible, and enforcing the 16-cut cap.

A high-level policy selects the cut budget and a low-level pointer policy orders candidates (Vinyals et al., 2015). If k cuts are selected from state S, their joint probability is

$$\pi_\theta(k, i_{1:k} \mid S) = \pi_\theta^H(k \mid S) \prod_{t=1}^{k} \pi_\theta^L(i_t \mid S, i_{1:t-1}). \tag{23}$$

Two workers collect independent SCIP trajectories in parallel; their batches then drive a synchronous central update of the shared policy with return

$R = -\text{PDI}$. The estimator

$$\widehat{\nabla_\theta J} = \frac{1}{B}\sum_{b=1}^{B}(R_b - \bar{R}_{\text{EMA}})\nabla_\theta \log \pi_\theta(\tau_b), \qquad \bar{R}_{\text{EMA}} \leftarrow 0.9\bar{R}_{\text{EMA}} + 0.1\bar{R} \quad (24)$$

uses an exponential-moving-average return baseline, not an independently learned critic. It is therefore hierarchical REINFORCE (Williams, 1992) with A3C-inspired parallel trajectory collection, not the asynchronous actor–critic algorithm of Mnih et al. (2016). Training uses five training instances and five seeds. For each seed, two workers collect eight 30-s/1024-MB trajectories per epoch for 60 epochs; Adam learning rates are 10^{-4} and 5×10^{-4} for the low- and high-level policies. For each family and seed, evaluation uses the predetermined terminal-epoch (epoch 60) parameter set; the candidate count is one. Eligibility is checked only from the training path, seed, PDI reward, feature dimension, and epoch; no benchmark- or test-based parameter selection is performed.

At inference, width-three beam search ranks partial sequences by cumulative joint log probability without length normalization, rather than committing to the first locally best cut (Choo et al., 2022). Role-submodular completion then adds candidates that balance policy score, immediate cut quality, role coverage, and representation of the remaining pool (Nemhauser et al., 1978). The mechanisms are complementary: reconstruction supplies meaningful diversity; beam decoding explores alternative orders; completion prevents the final set from collapsing onto one variable role.

4.4. Exactness and computational scope

RDMCT-HBS ranks only cuts already generated and certified as valid by SCIP (Achterberg, 2009; Bestuzheva et al., 2023). SPBS adds only a solver-checked incumbent. Consequently, neither component changes the objective, constraints, integrality requirements, or lower-bound logic. The method can improve anytime behavior while preserving the same mathematical meaning of feasibility and optimality as default SCIP. The supplementary material documents the reproducibility protocol.

5. Computational results and analysis

5.1. Evidence structure

The study uses seven complementary evidence sets. Four exactly solved cases check internal model consistency. A 2700-run nominal matrix compares

seven methods over 20 manufacturing instances and independent solver and training realizations. A 960-run matched experiment compares downstream SCIP solves with a precomputed SPBS incumbent against solves without that incumbent. A 1500-run study compares three equal-total-budget workflows: the two refinement profiles use 540 s for the primary solve and up to 60 s for refinement, whereas the no-refinement profile uses all 600 s for the primary solve. A 300-run factorial design estimates associations of wafer count, recipe composition, and process scale with difficulty. An 80-run one-factor study assesses solvability under parameter perturbations. Finally, 729 runs on nine shifted 40–64-wafer configurations assess larger-instance feasible-schedule delivery. All saved incumbents supporting reported conclusions pass SCIP's check and a post-solve feasibility check against the same serialized MILP.

PDI is the primary anytime metric because it integrates SCIP's normalized primal–dual gap over the common horizon; lower is better. Terminal gap is the normalized bound gap at termination. Reported solution time is the sum of SCIP-reported primary and conditional-refinement solve times; it excludes model loading, learned-parameter loading, solution serialization, instance construction, and offline SPBS generation. Repetitions are aggregated within manufacturing instances before paired bootstrap intervals, Wilcoxon tests, and Holm adjustment are calculated. This prevents repeated random seeds from being treated as independent manufacturing evidence (Gleixner et al., 2021). Hardware, budgets, hyperparameters, complete matrix definitions, and validation tolerances are reported in the supplement.

5.2. Exact solves and post-solve checks establish internal consistency

Table 3 reports four exact pure and mixed-flow cases. Both stages reach zero gap, and post-solve checking confirms every represented route, resource, cleaning, residence, and dummy-allocation row in the same serialized MILP. These checks are independent of the learning policy but do not constitute independent formulation or field validation.

The pure cases exercise the distinct rotary logic of each recipe, while mixed cases require both recipe structures, two load-lock paths, shared robots, and dummy reuse to coexist in one feasible trajectory. The 16-product case also exercises the coupled activation and resource-order constraints targeted by RDMCT-HBS.

Table 3: Exactly solved cases used for internal formulation-consistency checks.

Configuration	Products	$C_{\max}$	Route-total wait	Route-maximum wait	Time (s)
Pure 4×1	4	458	380	76	0.009
Pure 2×2	4	584	112	28	0.039
Mixed flow	8	834	506	76	0.458
Mixed flow	16	1170	1550	99	24.390

5.3. RDMCT-HBS attains the lowest observed nominal mean PDI and time

Table 4 gives the full nominal comparison. RDMCT-HBS has the lowest mean PDI and the shortest mean solution time among all seven methods. Its mean PDI is 1.41% below adapted HEM and 1.97% below default SCIP. Every one of its 500 runs returns a feasible incumbent.

All seven conditions receive the same precomputed, solver-checked SPBS incumbent, time limit, memory limit, and solver seeds. The comparison therefore isolates cut-selection and decoding choices rather than the total SPBS effect, which is identified separately below. SCIP denotes its default cut selector under this shared start; adaptive cut selection (ACS) uses a four-score SCIP rule selected on four validation instances by a simplex grid with step 0.25, using mean validation performance. HEM-13D and HEM-13D+Beam share the original 13-feature policy and differ only in decoding. The 23D-only condition uses reconstructed features without structural reranking; Structure+Greedy adds role-based completion; and RDMCT-HBS combines that structure with beam decoding.

Table 4: Nominal comparison under a common SPBS start; statistics are first averaged within 20 manufacturing instances.

Method	Mean PDI	Mean gap (%)	Mean time (s)	Optimal (%)
SCIP default selector	16919.01	**119.54**	226.05	**64.0**
Adaptive cut selection (ACS)	16854.87	127.18	222.34	62.0
HEM-13D	16823.18	127.24	219.65	62.4
HEM-13D+Beam	16829.59	127.19	219.95	62.4
23D Features Only	16620.91	125.33	219.42	62.6
Structure+Greedy	16598.11	126.03	218.44	63.2
RDMCT-HBS	**16585.70**	126.01	**218.32**	63.2

The descriptive pattern across methods is consistent with complementary module roles. Relative to HEM-13D, the 23D-only condition has the largest

numerical decrease in mean PDI; Structure+Greedy and then RDMCT-HBS have successively lower means, while beam decoding alone slightly worsens HEM-13D. Because these component contrasts are not statistically significant after adjustment, they do not identify a causal contribution from any individual module. They instead motivate testing beam decoding jointly with algebraic role information.

SCIP has a marginally better final gap and optimal rate, whereas RDMCT-HBS leads PDI and elapsed time. This is an interpretable trade-off rather than a contradiction: SCIP's default selector records stronger terminal proof metrics, while RDMCT-HBS records lower observed mean anytime and elapsed-time metrics. The experiment does not identify which cut-level mechanism causes that pattern. For time-constrained industrial use, where an incumbent and its quality during the available horizon matter, RDMCT-HBS has the lowest observed nominal mean PDI and time. The instance-first mean PDI improvements over HEM and SCIP are 237.49 (95% bootstrap CI $[-446.71, 846.26]$) and 333.31 (CI $[-382.16, 999.70]$), respectively; both Holm-adjusted tests have $p = 1$. Thus the nominal ranking is an observed mean ordering over this matrix, whereas the statistically supported matched contrasts are the SPBS and equal-budget workflow comparisons reported next.

5.4. Precomputed SPBS improves downstream solver-phase outcomes

The matched SPBS experiment gives the clearest statistically supported solver-phase contrast. As Table 5 shows, supplying a precomputed SPBS incumbent raises incumbent availability from 51.67% to 100% and reduces mean PDI by 26.19%. The contrasts remain significant after Holm adjustment ($p = 2.55 \times 10^{-5}$ and $p = 3.46 \times 10^{-8}$, respectively).

Table 5: Downstream SCIP comparison over 48 instances and ten matched seeds; SPBS construction time is excluded.

Warm start	Runs	Incumbent (%)	Optimal (%)	Mean PDI
Precomputed SPBS	480	**100.00**	54.58	**20278.78**
None	480	51.67	51.67	27473.05

Without a supplied start, nearly half of the runs end before a complete schedule is available. The observed PDI and availability differences are consistent with SCIP receiving a feasible incumbent at the beginning of its search

and then improving or exploiting it. Because construction is performed offline and is not charged, this experiment estimates the downstream value of loading SPBS; it does not establish a net end-to-end time advantage after incumbent construction.

5.5. Equal-budget refinement workflows improve operational metrics

Makespan verification alone cannot establish the value of refinement. We therefore reconstruct route waits directly from product event times and cadence directly from active chamber starts. This avoids using objective slack variables as performance measures and makes the metric identical across profiles. Table 6 compares workflows under the same 600-s total budget. The full and quadratic profiles allocate 540 s to the primary solve and up to 60 s to refinement, whereas the no-refinement profile allocates all 600 s to the primary solve. The contrast is consequently not a pure stage-two intervention conditional on an identical stage-one incumbent.

Table 6: Operational metrics for equal-600-s-budget refinement workflows.

Profile	Route-total wait	Route-maximum wait	Cadence CV	Cadence deviation
Full two-stage model	2596.01	296.33	**0.130**	**22.04**
Quadratic objective (linear off)	**2469.47**	**293.27**	0.134	60.18
No second stage	5398.60	952.57	0.216	370.41

Relative to the no-refinement workflow, the full two-stage workflow has 51.91% lower route-total wait, 68.89% lower maximum route wait, 39.70% lower cadence CV, and 94.05% lower cadence deviation; all four matched contrasts remain significant after Holm adjustment. These are large workflow-level differences. They support the operational usefulness of the prespecified 540+60-s configuration, while differences in the primary incumbents induced by the budget split may contribute to the observed magnitudes.

The quadratic profile has slightly lower route-wait means, but the full linear profile has 63.38% lower cadence deviation in the matched comparison. Thus the linear 540+60-s workflow records the strongest observed cadence regularity while retaining route waits far below the 600-s no-refinement workflow. Because the two refinement profiles use the same split, their contrast is more specific to objective profile than the comparison against no refinement.

5.6. Difficulty rises with scale and mixed-flow interaction

Table 7 shows a sharp observed complexity transition. All 8- and 16-wafer runs prove optimality, but the rate falls to 36% at 24 wafers and 20%

at 32 wafers. Mean PDI increases from 28.13 and 1876.54 to 31155.94 and 32959.89, respectively.

Table 7: Computational transition by wafer count.

Wafers	Runs	Optimal (%)	Mean PDI	Mean time (s)
8	75	100	28.13	0.34
16	75	100	1876.54	33.86
24	75	36	31155.94	408.55
32	75	20	32959.89	433.20

In the prespecified fitted factorial model ($R^2 = 0.998$ for PDI), the largest estimated terms are interactions between wafer count and recipe composition. At the 1:1 mix, the additional PDI interaction estimates are 37729.48 and 40788.63 for 24 and 32 wafers. Every prespecified processing-scale main and interaction interval includes zero. Within this 0.9–1.1 scale range, the design therefore detects no processing-scale association; it does not establish that processing duration has no effect outside the tested range. The observed interaction pattern is consistent with greater combinatorial coupling in mixed, larger instances. The complete coefficient table and confidence intervals are retained in the reproducibility artifacts; the main text reports the prespecified effects required for interpretation.

5.7. Sensitivity study retains feasible-schedule delivery

All 80 one-factor sensitivity runs return solver-checked incumbents. The lowest mean PDI occurs with a longer cleaning interval (28899.41), whereas slower motion produces the highest mean PDI (42478.32). The pattern is compatible with fewer dummy-only interruptions at longer cleaning intervals and tighter shared-robot timing under slower motion, although the one-factor study does not identify either mechanism causally. Feasible-schedule delivery is retained across every prescribed perturbation.

5.8. OOD stress test retains feasible-schedule delivery

The distribution-shift suite enlarges the batch to 40–64 wafers and changes the product-mixture composition. Every one of the 729 runs returns a saved incumbent that passes SCIP and post-solve checks against the same MILP. Because every run uses the complete search horizon, PDI and terminal gap provide the discriminating measures in Table 8.

Table 8: OOD stress test; statistics are first averaged within nine manufacturing instances.

Method	Runs	Incumbent (%)	Mean PDI	Mean gap (%)
SCIP	27	100	73670.99	171.35
Adaptive cut selection	27	100	**70915.54**	**160.78**
HEM-13D	135	100	73003.59	165.88
HEM-13D+Beam	135	100	73165.58	167.04
23D Features Only	135	100	74187.22	169.96
Structure+Greedy	135	100	73507.76	168.54
RDMCT-HBS	135	100	73919.15	170.02

Adaptive cut selection records the lowest OOD PDI, while RDMCT-HBS has an observed mean terminal gap 0.78% below default SCIP. No OOD pairwise contrast remains significant after global Holm adjustment. The complete outcome separates two forms of evidence: RDMCT-HBS leads observed mean anytime search in the nominal production range, and the shared SPBS–MILP–SCIP pipeline delivers a solver-checked feasible schedule in every tested larger shifted run. All seven methods use the full search horizon and none proves OOD optimality; global bound closure therefore remains unresolved at this scale.

5.9. Industrial implications

Three implications follow from the evidence. First, the prespecified two-stage workflows have substantially better wait and cadence metrics than the equal-total-budget no-refinement workflow, subject to the 540+60 versus 600-s primary-search confound. Second, supplying a precomputed application-feasible incumbent gives the largest statistically supported computational contrast measured here, although its offline construction cost is outside that comparison. Third, the numerical component pattern motivates representing a cut's algebraic variable domains. The observed sign reversal of the beam contrast is consistent with coupling between beam search and role information, but neither the individual module contrasts nor their interaction are statistically established.

The explicit state, resource, and timing layers also provide a possible foundation for online arrivals, stochastic processing, and module-failure recourse. The present finite-batch evidence shows that, on the tested canonical-slot instances, equal-budget two-stage workflows improve the reported operational

metrics and RDMCT-HBS attains the lowest observed nominal mean PDI and time among the compared methods without a significant pairwise advantage.

6. Conclusions

This paper develops an integrated model-and-algorithm framework for a class of rotating-chamber semiconductor cluster tools. The source interface separates external material topology from a common internal configuration–resource–timing model: single-source operation is the base specialization, and the dual-source mixed-flow configuration is the evaluated stress case. The two-stage MILP represents product wafers, reusable vacuum-zone dummy wafers, mixed rotary recipes, cleaning, shared robots, load locks, and residence restrictions under fixed canonical same-mode pairing and chamber–side slots. Its first stage optimizes end-to-end makespan; its second stage retains the discrete production plan and refines its timing quality.

Four exactly solved cases and post-solve checks establish internal consistency for the tested instances. Relative to the 600-s no-refinement workflow, the 540+60-s two-stage workflow has 51.91%, 68.89%, 39.70%, and 94.05% lower route-total wait, maximum route wait, cadence variation, and cadence deviation; this equal-budget comparison does not isolate refinement conditional on an identical primary incumbent. RDMCT-HBS records the lowest observed mean PDI and solution time in the 2700-run seven-method comparison, but its pairwise advantages are not significant after adjustment. Supplying precomputed SPBS is associated with 26.19% lower downstream-solve PDI and 100% incumbent availability; construction time is excluded. The factorial study identifies the largest estimated PDI terms as scale–recipe-mixture interactions within the tested design. All 729 OOD runs produce solver-checked incumbents on nine larger shifted configurations, although none proves optimality.

The results support the operational usefulness of the prespecified two-stage workflow and document the lowest observed nominal mean anytime performance for the structure-aware learning-enhanced branch-and-cut method in this comparison. The scope of inference is the tested canonical-slot instances, budgets, and distributions.

CRediT authorship contribution statement

Lizhen Shao and Lincong Wei jointly contributed to conceptualization, methodology, software, validation, formal analysis, investigation, visualization, and manuscript preparation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The data, mathematical instances, trained models, and source code supporting the findings are available from the corresponding author on reasonable request. A permanent public archive will be cited in the version of record.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI ChatGPT and Codex for language editing, LaTeX consistency checking, literature-metadata verification, software debugging, experiment-workflow scripting, code review, and drafting or refining analysis text. It was not used to generate experimental observations, alter solver output, select favorable runs, or make authorship decisions. The authors checked and approved all mathematical statements, citations, code changes, statistical interpretations, and final text, and take full responsibility for the content of the publication.

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